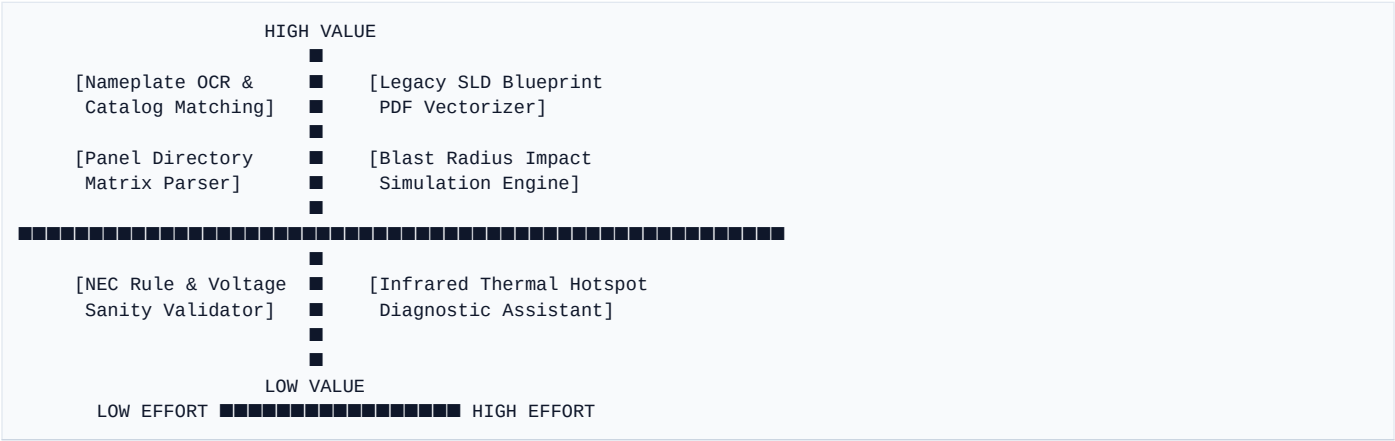


JAMES Electrical Digital Twin: AI Features & Capabilities Roadmap

1. Executive Summary

The **JAMES Electrical Digital Twin** platform bridges physical field survey workflows with authoritative electrical engineering data models. By integrating multimodal AI and domain-grounded intelligence, JAMES transforms from a manual survey tool into an **autonomous electrical engineering copilot**—dramatically reducing field data collection time, eliminating data transcription errors, verifying National Electrical Code (NEC) compliance, and enabling predictive facility operational analysis.

2. Strategic AI Feature Matrix



3. Core Feature Pillars

Pillar I: Multimodal Vision & Field Data Acceleration (Field Survey Superpowers)

1. Nameplate OCR & Catalog Matching (Gemini Multimodal)

- **Problem:** Field electricians spend considerable time typing model numbers, AIC ratings, serial numbers, and full-load amps from dirty or inaccessible nameplates.
- **Solution:** Electrician snaps a photo of an equipment nameplate (Eaton, Square D, Siemens, GE, Baldor, etc.).
- **AI Execution:**
 - Extracts Volts, Phase, Poles, FLA, NEMA type, Frame size, AIC rating, and Serial Number.
 - Matches raw string extractions to normalized SKUs in the Master Vendor Catalog.
 - Auto-populates the survey form and locks in verified engineering tolerances.

2. Panel Directory Photo-to-Schedule Transcriber

- **Problem:** Entering 42 or 84 circuit breaker rows with handwritten or faded directory labels takes 15–20 minutes per panel.
- **Solution:** Take a high-resolution photo of the paper directory glued inside the panel door.
- **AI Execution:**
 - Segregates Odd (Left) vs. Even (Right) slot columns.
 - Detects bracketed multi-pole ties (2P, 3P circuits).

- Transcribes text (e.g. AHU-1 Supply Fan, Lab Receptacles RM 104, Spare).
- Instantly populates the interactive breaker matrix.

3. Legacy Single-Line Diagram (SLD) & Blueprint Ingestion

- **Problem:** Facilities have 30 years of scanned PDF blueprints and CAD drawings with no digital twin database.
- **Solution:** Upload PDF single-line drawings into JAMES.
- **AI Execution:**
 - Recognizes standard electrical symbols (Transformers, Main Switchboards, ATS, Breakers, Panelboards, Generators, VFDs).
 - Parses upstream/downstream connection lines and conductor sizes.
 - Generates a fully linked `model.db` / `model.json` graph structure ready for verification.

4. Infrared (IR) Thermal Inspection Diagnostic Assistant

- **Problem:** Thermographers take FLIR images during PM routines, but report generation and severity classification is manual.
 - **Solution:** Upload radiometric IR photos directly to an asset's survey record.
 - **AI Execution:**
 - Detects hot-spot temperature deltas (ΔT) between Phase A, B, and C bus stabs.
 - Automatically flags loose lugs, overloaded phases, or deteriorating breaker contacts according to NETA/NFPA 70B standards.
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Pillar II: Engineering Intelligence & Automated Topology Validation

1. Topology Sanity & Voltage Mismatch Sentinel

- Real-time automated verification engine that audits the graph on every connection update:
- **Voltage Incompatibility:** Flags if a 208Y/120V panel is fed from a 480V switchboard without an intermediate step-down transformer (XFMR).
- **Ampacity & OCPD Violation:** Warns if an upstream 400A breaker feeds a panel with a 225A bus rating without proper secondary protection.
- **Phase Rotation Discontinuity:** Ensures 3-phase source chains do not improperly feed single-phase designated downstream loads without tagging.

2. Automated Arc Flash & NFPA 70E Pre-Computation

- Estimates available short-circuit current (I_{sc}) based on utility transformer size and impedance ($Z\%$).
- Computes incident energy (cal/cm^2) at each panel bus.
- Recommends required PPE category (Category 1–4) and prints standard compliant Arc Flash labels directly from JAMES.

3. Conductor Sizing & Conduit Fill Calculation

- Given load FLA, distance, and conduit type, automatically calculates minimum copper conductor size, voltage drop ($< 3\%$ feeder, $< 5\%$ total), and conduit fill percentage (NEC Chapter 9).
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Pillar III: Conversational Digital Twin ("Chat with Your Facility")

1. De-energization & "Blast Radius" Simulation

- Natural language query interface for facility managers and maintenance staff:
- *Query*: "What happens if I de-energize Feeder Space #7 on MDP-1 for 2 hours?"
- *AI Response*: Traverses the graph downstream and outputs:
- Affected Panels: LP-1A
- Critical Loads Dropped: Central Plant Chiller CH-1, Chilled Water Pump P-1
- Affected Facility Areas: Level 1 Cleanrooms, Penthouse Mechanical Room
- Standby Power Availability: Checks if ATS-1 can transfer critical loads to GEN-1.

2. Predictive Capacity & Space Utilization Auditing

- *Query*: "Where can I add a new 60A 480V 3-phase autoclaving machine on Floor 2?"
- *AI Response*: Scans all panels in Electric Room 201, evaluates unused physical slots/spaces, calculates remaining transformer kVA head-room, and suggests:
- MCC-1 Buckets 13, 15, 17 (Available spare capacity: 120A, 480V 3P).

3. Compliance & Obsolescence Search

- *Query*: "List all circuit breakers in the facility with AIC ratings below 25kA that are older than 15 years."
- *AI Response*: Returns a structured table of candidate assets for capital replacement planning.

Pillar IV: Master Vendor Catalog AI Enrichment

1. Autonomous Spec Sheet Extraction

- Crawls manufacturer technical datasheets (PDF cut-sheets from Eaton, Square D, Siemens, ABB, Generac) to auto-extract dimensions, trip curve parameters, interrupt ratings, and CAD port coordinates.

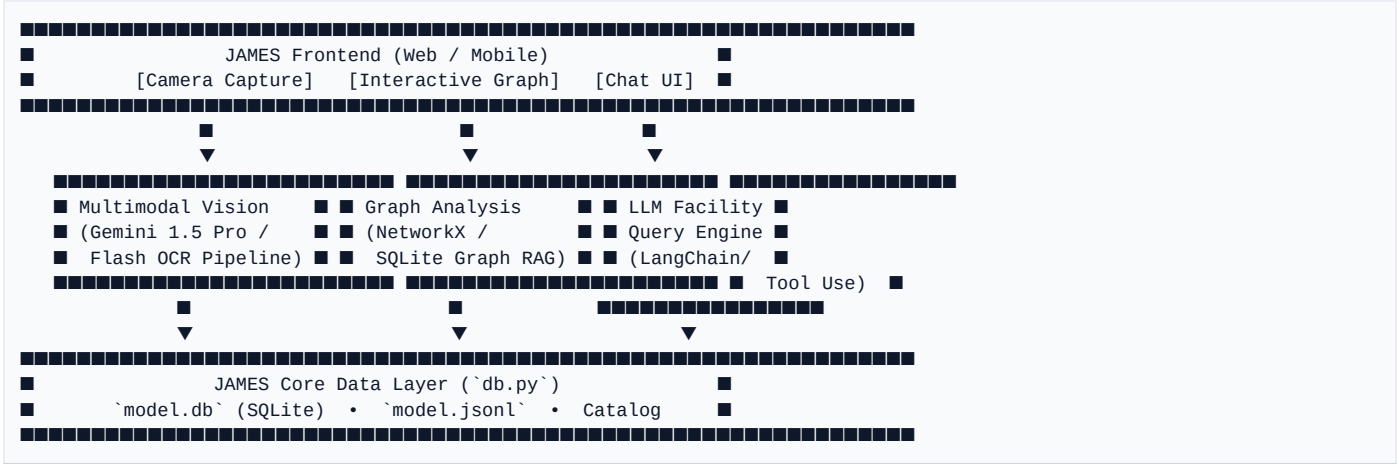
2. Cross-Vendor Equivalent & Replacement Matcher

- Suggests drop-in catalog replacements when legacy equipment is marked as obsolete:
- *Example*: Recommends contemporary *Eaton Power Defense Frame 2* equivalent for an obsolete *Westinghouse / Cutler-Hammer Series C* molded-case breaker.

4. Implementation Phasing

Phase	Milestone / Features	Target Timeline
Phase 1: Quick Wins	• Nameplate Photo OCR & Catalog Matching • Panel Directory Door Card Transcriber	Month 1–2
Phase 2: Graph Safety	• Live Topology & Voltage Sanity Validator • Automated Conductor & Conduit Sizing	Month 3–4
Phase 3: Digital Twin Chat	• Natural Language Graph Traversal (Blast Radius Queries) • Capacity & Spare Slot Search Assistant	Month 5–6
Phase 4: Advanced Vision	• Legacy PDF Single-Line Blueprint Vectorizer • FLIR Infrared Thermal Anomaly Classifier	Month 7+

5. Technical Architecture for AI Services



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